

**Nirma University  
Institute of Technology**



**Course Code :- 6CS103CC26**  
**Course Name :- Operating System**

**Practical - 4**

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# Assignment Task 2: Directory Management

## Objective

The purpose of this assignment is to develop a shell script that demonstrates proficiency in directory manipulation commands and interactive user input handling. The script is designed to perform four distinct tasks: directory navigation, directory listing, directory creation, and directory modification.

## Implementation Details

The script begins by implementing an interactive menu driven by a case control structure. It accepts a user's selection and routes execution to the appropriate directory management function, ensuring a clean and modular interface for all operations.

Following the menu dispatch, the script supports directory navigation using both absolute and relative pathnames. The user provides a path string, which is passed to the `cd` command to change the working directory. The current location is then confirmed using the `pwd` command to verify successful navigation.

Next, the script lists the contents of the current directory by invoking the `ls` command. This provides a straightforward view of all files and subdirectories within the active working directory.

Finally, the script handles directory creation and modification. The `mkdir` command is used to create a new directory with a user-supplied name, while the `mv` command renames an existing directory by accepting both the old and new names as inputs. Each section of the script is modular to ensure clarity and ease of execution.

## Source Code

```
#!/bin/bash
echo "1. Navigate Directory"
echo "2. List Contents"
echo "3. Create Directory"
echo "4. Modify"
echo "5. Exit"
read -p "Select an option (1-5): " choice

case $choice in
    1)
        read -p "Enter Path: " path
        cd "$path"
        echo "Current path: $(pwd)"
        ;;
```

```

2)
    echo "Directory Contents"
    ls
    ;;
3)
    read -p "Enter new directory name: " dname
    mkdir "$dname"
    echo "Directory created."
    ;;
4)
    read -p "Enter your old Directory name:" d1
    read -p "Enter your new Directory name:" d2
    mv "$d1" "$d2"
    echo "Changed Directory Name"
    ;;
5)
    echo "Exit"
    exit 0
    ;;
*)
    echo "Invalid"
    ;;
esac

```

```

● nirma@N503-136:~/Downloads/new$ bash 4.sh
1. Navigate Directory
2. List Contents
3. Create Directory
4. Modify
5. Exit
Select an option (1-5): 1
Enter Path: hello
Current path: /home/nirma/Downloads/new/hello

```

```

● nirma@N503-136:~/Downloads/new$ bash 4.sh
1. Navigate Directory
2. List Contents
3. Create Directory
4. Modify
5. Exit
Select an option (1-5): 2
Directory Contents
4.sh demo.txt hello rajvi world

```

```
● nirma@N503-136:~/Downloads/new$ bash 4.sh
1. Navigate Directory
2. List Contents
3. Create Directory
4. Modify
5. Exit
Select an option (1-5): 3
Enter new directory name: vinay
Directory created.
```

```
● nirma@N503-136:~/Downloads/new$ bash 4.sh
1. Navigate Directory
2. List Contents
3. Create Directory
4. Modify
5. Exit
Select an option (1-5): 4
Enter your old Directory name:hello
Enter your new Directory name:demo
Changed Directory Name
```

## Conclusion

This script demonstrates the fundamental directory management operations in a Unix/Linux shell environment. By combining a case-based menu with standard commands such as `cd`, `pwd`, `ls`, `mkdir`, and `mv`, the script provides a practical, interactive interface for navigating, listing, creating, and modifying directories. The modular structure ensures that each operation is independent and easily extendable, reinforcing core shell scripting concepts like user input handling and conditional branching.